

15.4 Applications of Double Integrals

Prerequisite information for this section: There is no prerequisite assignment for this section.

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Center of mass (of a lamina):** [see discussion before Figure 4 in section 15.4.]

The center of mass $(\bar{x}, \bar{y}) = (\frac{M_x}{M}, \frac{M_y}{M})$, where

$$M_i = \iint_D \rho(x, y) dA$$

- **Moment of inertia of a particle of mass m :** [See discussion before Box # 6 in section 15.4.]

$$mr^2$$

- **Probability density function (p.d.f.) for a random variable X :** [See the *first* paragraph under the heading “Probability”, and use this information to give the **two characteristics** of the p.d.f.]

The probability density function is some f where $f(x) \geq 0 \quad \forall x \in \mathbb{R}$,

$\int_{-\infty}^{\infty} f(x) dx = 1$, and the probability that some $X \in [a, b]$ is given by

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Topics to learn in this section:

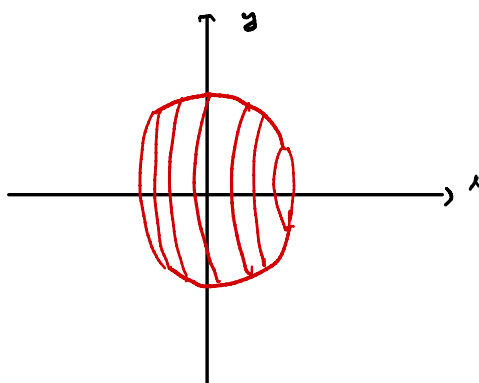
1. Density, mass, and centers of mass (for an engineering- or physics-oriented course)
2. Probability and expected values (for a course oriented toward biology or the social sciences)

15.5 Surface Area

Prerequisite information for this section:

In Calculus 2, you found the surface area of shapes created by revolving the graph of a function about the x - or y -axis. Use this technique to

1. Sketch a graph of the curve $y = \sqrt{9 - x^2}$, for $-1 \leq x \leq 2$, and of the surface obtained by rotating this arc about the x -axis.



2. Use methods from Calculus 2 (as seen in section 8.2 of your text, boxed formula #4 or #5 and similar to Example 1) to calculate the surface area of this surface.

$$\begin{aligned}
 SA &= \int_{-1}^2 2\pi \sqrt{9-x^2} \sqrt{1 + \frac{x^2}{9-x^2}} dx = 2\pi \int_{-1}^2 \sqrt{9-x^2 + x^2} dx \\
 &= 2\pi \int_{-1}^2 3 dx = 6\pi (2-(-1)) = 18\pi
 \end{aligned}$$

$$\frac{d}{dx} \sqrt{9-x^2} = \frac{-2x}{2\sqrt{9-x^2}} = -\frac{x}{\sqrt{9-x^2}}$$

Definitions in this section: There are no new definitions in this section.

Topics to learn in this section:

1. The derivation of the formula for the area ΔT_{ij} used to approximate the surface area over a small region.
2. Various techniques and difficulties associated with surface area integrals.

15.6 Triple Integrals

Prerequisite information for this section: Evaluate each integral below, showing your work. This is a review of some basic integration techniques from Calculus 1 and 2. Feel free to use another sheet of paper if you need more space.

$$\begin{aligned}
 1. \int_0^\pi \frac{\sin \sqrt{x}}{\sqrt{x}} dx & \quad u = \sqrt{x} \Rightarrow du = \frac{dx}{2\sqrt{x}} \\
 & = \int_0^{\sqrt{\pi}} \frac{\sin(u)}{\sqrt{x}} 2\sqrt{x} du = -2\cos(u) \Big|_0^{\sqrt{\pi}} = -2\cos(\sqrt{\pi}) + 2
 \end{aligned}$$

$$\begin{aligned}
 2. \int_0^{1/3} x \cos(\pi x) dx & = \frac{\sin(\pi x)}{\pi} \Big|_0^{1/3} - \int_0^{1/3} \frac{\sin(\pi x)}{\pi} dx = \frac{\sin(\pi/3)}{\pi} - \frac{\cos(\pi x)}{\pi^2} \Big|_0^{1/3} \\
 x=u \quad \cos(\pi x) &= du \\
 1=du \quad \frac{\sin(\pi x)}{\pi} &= v \\
 & = \frac{\sqrt{3}}{2\pi} - \frac{1}{2\pi^2}
 \end{aligned}$$

$$3. \int_1^2 \frac{e^{1/x}}{x^2} dx = \int_1^2 x^{-2} e^{x^{-1}} dx = \int -e^u du = -e^u \Big|_1^2 = e^1 - e^{1/2}$$

$$u = x^{-1}$$

$$du = -x^{-2} dx$$

Definitions in this section: Extend the meanings of the terms below (defined for laminas in Section 15.4) to three dimensional space, using the discussion beginning in the paragraph before Box # 13 and ending with the paragraph after Box #15 in section 15.6.

- Mass of a 3-dimensional object:

Let an object occupying a region $E \subset \mathbb{R}^3$ have a density function $\rho(x, y, z)$.

The mass is defined as

$$m = \iiint_E \rho(x, y, z) dV$$

- Center of mass (centroid) of a 3-dimensional object:

The center of mass of an object is given as

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{M_{yz}}{m}, \frac{M_{xz}}{m}, \frac{M_{xy}}{m} \right)$$

Topics to learn in this section:

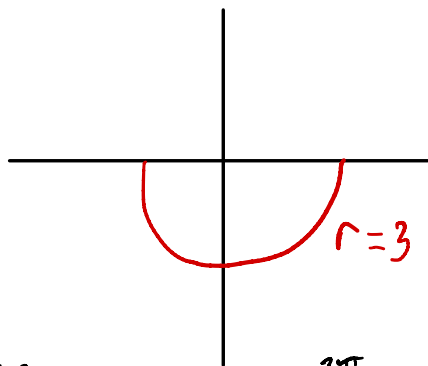
1. The basic definition of a triple integral (at first, in rectangular coordinates).
2. The various types of volume domain, and how to set up the volume integral based on each of them.
3. Changing the order of integration in triple integrals.

15.7 Triple Integrals in Cylindrical Coordinates

Prerequisite information for this section: Consider the integral below and complete each of the following:

$$\int_0^3 \int_{-\sqrt{9-y^2}}^0 x^2 y dx dy$$

- Sketch the region of integration in the xy -plane.
- Convert the integral to polar coordinates.
- Evaluate the integral.



$$\int_{\pi}^{2\pi} \int_0^3 r^3 \cos^2 \theta dr d\theta = \int_{\pi}^{2\pi} \frac{3^4}{4} \cos^2 \theta d\theta = \frac{81}{4} \int_{\pi}^{2\pi} \frac{1 + \cos(2\theta)}{2} d\theta$$

$$= 81 \left[\frac{1}{2} (2\pi - \pi) + \frac{\cos(4\pi) - \cos(2\pi)}{2} \right] = 81 \cdot \frac{\pi}{2} = \frac{81\pi}{2}$$

Definitions in this section: There are no new definitions in Section 15.7.

Topics to learn in this section:

1. The cylindrical coordinate system as an extension of polar coordinates in $\mathcal{R}^2$.
2. The basic shapes of cylindrical solids.
3. The idea that the cylindrical coordinate system can be used to simplify equations and volume integrals of certain three-dimensional surfaces and solids.